

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The proposed technical architecture uses ServiceNow as the central IT Service Management platform. It connects requesters, approvers, network fulfillment teams and administrators through a standardized service catalog, automated workflow, business logic, data storage, notifications, reporting and optional external integrations.

Architecture Flow: Users → ServiceNow Employee Center / Service Portal → Network Service Catalog → Request Forms & Validation → Flow Designer → Approval & Assignment → Network Fulfillment → Notifications & Tracking → Reporting / Audit

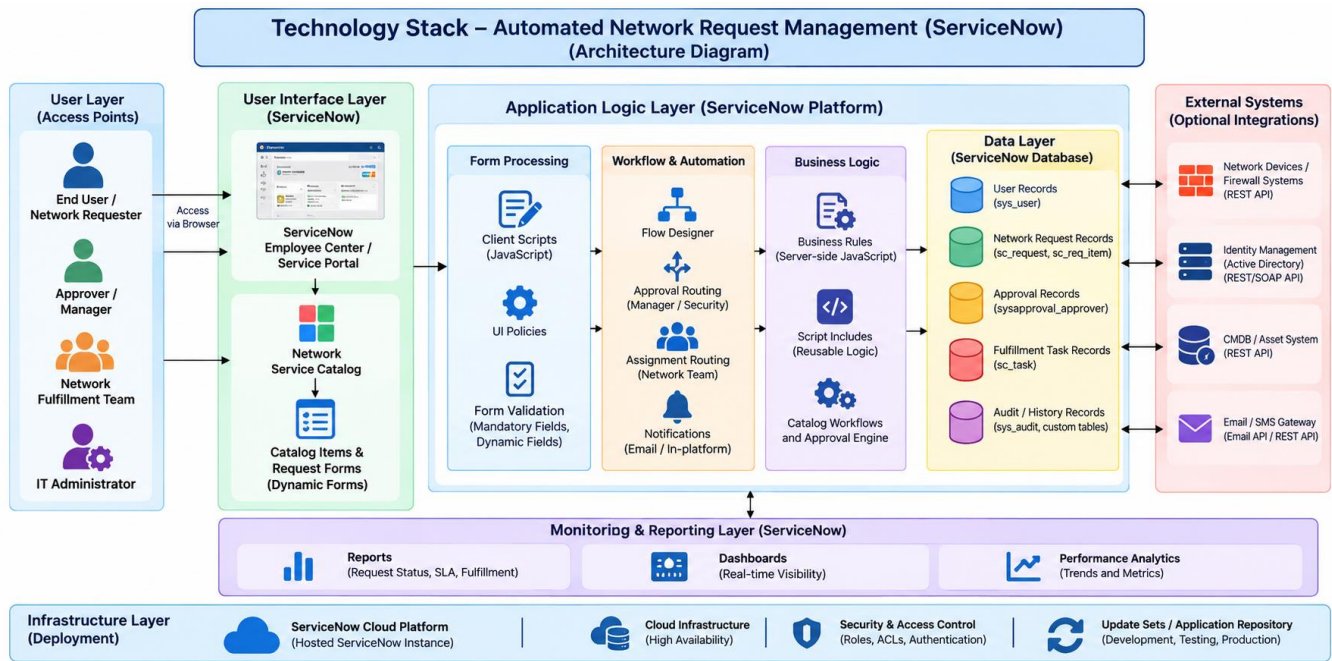


Figure 1: Proposed Technical Architecture – Automated Network Request Management

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interface for submitting and tracking network requests.	ServiceNow Employee Center / Service Portal, Service Catalog
2	Application Logic-1	Client-side form behavior, mandatory fields, validation and dynamic fields.	Client Scripts, UI Policies, JavaScript
3	Application Logic-2	Server-side processing, business rules and reusable application logic.	Business Rules, Script Includes, JavaScript
4	Application Logic-3	Workflow automation, approvals, assignment, tasks and notifications.	Flow Designer, Approval Engine, ServiceNow Notifications
5	Database	Stores users, requests, approvals, tasks, status and history.	ServiceNow Platform Database / ServiceNow Tables
6	Cloud Database	Central platform data storage; no separate external database is required for the core solution.	ServiceNow Platform Database
7	File Storage	Stores supporting attachments when required by network requests.	ServiceNow Attachment Storage
8	External API-1	Optional integration with network devices or firewall systems for fulfillment.	REST API / IntegrationHub
9	External API-2	Optional integration with identity, CMDB/asset or enterprise systems.	REST / SOAP APIs / IntegrationHub
10	Machine Learning Model	Not required because the proposed project focuses on workflow and IT service automation.	N/A
11	Infrastructure (Server / Cloud)	Hosts the application and platform	ServiceNow Cloud Platform /

		services for authorized users.	ServiceNow Instance
--	--	--------------------------------	---------------------

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The core solution is built on the ServiceNow platform and does not require a separate open-source application framework.	JavaScript / ServiceNow Platform APIs
2	Security Implementations	Role-based access, authentication, authorization, ACLs and audit logging protect request and administrative data.	ServiceNow Roles, ACLs, Authentication, Authorization, Audit Logs, OWASP practices
3	Scalable Architecture	New request types, users, approval rules and fulfillment groups can be added without redesigning the complete solution.	Service Catalog, Flow Designer, reusable configuration and platform tables
4	Availability	The proposed solution is hosted on the ServiceNow cloud platform and is accessible to authorized users through a web browser.	ServiceNow Cloud Platform / platform availability capabilities
5	Performance	Standardized forms, validation, efficient queries and automated flows reduce unnecessary manual processing and workflow delays.	ServiceNow Platform, Client Scripts, Business Rules, Flow Designer

Technology Stack Summary

Layer	Selected Technology
User Interface	ServiceNow Employee Center / Service Portal
Service Catalog	ServiceNow Service Catalog
Form Logic	Client Scripts, UI Policies, JavaScript
Server Logic	Business Rules, Script Includes
Automation	Flow Designer
Approval	ServiceNow Approval Management
Database	ServiceNow Platform Database
Integration	REST/SOAP APIs, IntegrationHub
Security	Roles, ACLs, Authentication, Authorization
Reporting	Reports, Dashboards, Performance Analytics
Hosting	ServiceNow Cloud Platform

Note: External integrations shown in the architecture are optional extension points. The core project can operate using ServiceNow's native catalog, workflow, database, notification, security and reporting capabilities.

References

ServiceNow Documentation – <https://www.servicenow.com/docs/>
ServiceNow Developer – <https://developer.servicenow.com/>
C4 Model – <https://c4model.com/>
IBM Cloud Architecture – <https://www.ibm.com/cloud/architecture>
AWS Architecture – <https://aws.amazon.com/architecture>